

Target-Speaker Speech-to-Text in Overlapping Speech

ECE1508 – Applied Deep Learning

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The Problem: ASR Fails on Overlapping Speech

- Modern ASR systems like Whisper perform well on clean, single-speaker audio
- But in meetings, courtrooms, and media, speakers frequently overlap
- Standard ASR has no notion of which speaker to transcribe and it tries to transcribe everything, which causes errors.

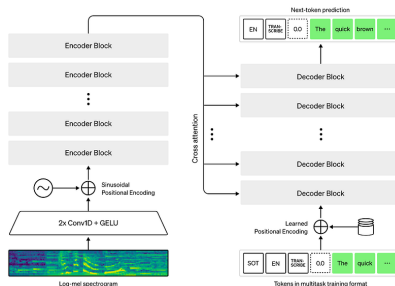
Goal: Given a short enrollment clip of a target speaker and a multi-speaker mixture, transcribe only the target speaker.

Why does this matter?

Accurate per-speaker transcription is needed in courtrooms, podcasts, debates, and meetings where speakers frequently talk over each other. It can also support full-duplex speech models that need labelled transcriptions of natural conversations with interruptions.

Our ASR Model: Whisper

We use OpenAI's Whisper as our ASR backbone. Whisper is a transformer-based encoder-decoder model trained on 680,000 hours of multilingual speech.



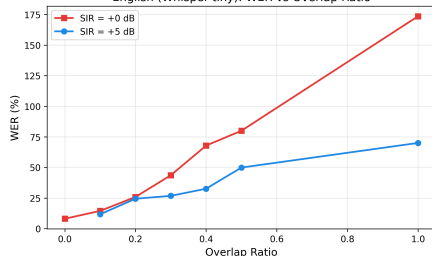
- English: We use Whisper-tiny (39M params) off the shelf
- Twi (Akan): Whisper does not support Twi natively, so we fine-tuned Whisper-small (244M params) on the WAXAL Akan dataset
- Both models serve as baselines to measure how overlap degrades transcription and whether the trend is observed across languages

Overlap Degrades ASR Across Languages

12,500 synthetic two-speaker mixtures per language at varying SIR levels and overlap ratios.

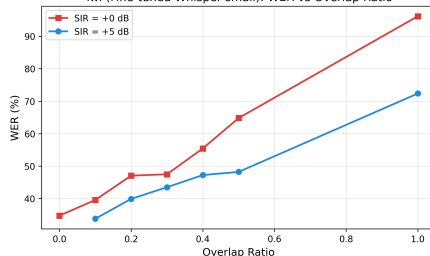
English (Whisper-tiny)

English (Whisper-tiny): WER vs Overlap Ratio



Two (fine-tuned Whisper-small)

Two (Fine-tuned Whisper-small): WER vs Overlap Ratio



- English clean baseline: 8.1% WER. At 0 dB SIR with full overlap, WER exceeds 100%.
- Two clean baseline: 34.7% WER. Same degradation pattern confirms this is language-agnostic.

Proposed Solution and Next Steps

Our approach: add a speaker-conditioned extraction stage before ASR. Whisper stays frozen.

- 1 Extract a speaker embedding from a short enrollment clip using a pretrained speaker verification model
- 2 A lightweight trained module uses the embedding to produce a time-frequency mask that isolates the target speaker from the mixture
- 3 Whisper (kept fixed) transcribes the cleaned signal

Remaining Work

- Implement speaker embedding extraction and conditioned masking module
- Train on synthetic mixtures, evaluate WER improvement over the unconditioned baseline
- Ablations: enrollment clip length, non-target leakage, cross-lingual transfer